

MONTHLY PROJECT REPORT

Handover Voice Summarizer

Web application (desktop + responsive mobile view) and companion native mobile app • Prepared by Muhammad • August 2026

2	371	~94%	6
codebases in scope	automated backend tests	backend test coverage	Docker Compose services

Executive Summary

Handover Voice Summarizer has grown from an early-stage FastAPI backend into a full production-shaped system: a care worker records a voice note, it is transcribed locally with Whisper, summarised into a structured handover by Gemini, and surfaced to managers in real time — with urgent cases triggering notifications. In the 30 days covered by this report (23 Jul – 21 Aug 2026, 21 commits across four branches), the backend gained real-time delivery, resident/staff assignment tracking, audit logging, and a full automated test suite wired into CI/CD; the existing React web app matured across both its desktop and responsive mobile layouts; and a brand-new, independent native mobile app (Expo/React Native) was stood up from scratch and is now feature-complete for day-to-day use, including in-app voice recording.

The system is no longer a prototype. It has a 6-service Docker Compose stack, a 371-test backend suite running in GitHub Actions with dependency and static-analysis security scanning, and three separate client surfaces (web desktop, web mobile, and native app) all talking to the same API. The main work remaining is deployment hardening (TLS, CORS lockdown, secrets, object storage) and finishing a couple of in-flight mobile screens — the core product is functionally done.

1. What the Product Does

- **Record:** a care worker opens a shift and records a voice note describing a resident's status, incidents, mood, and any follow-up actions needed.
- **Transcribe:** audio is transcribed on-device/on-server using a locally cached OpenAI Whisper model — no third-party audio upload.
- **Summarise:** the transcript is sent to Gemini (gemini-2.5-flash), which returns a structured handover: key events, medications, incidents, follow-up actions, resident mood, and an urgency level.
- **Notify:** handovers flagged urgent immediately alert managers by email and by an in-app real-time notification (WebSocket), so nothing urgent waits for someone to refresh a page.
- **Track:** managers assign care workers to residents, review the handover feed, manage shifts, and see a full audit trail of who did what and when.

Two roles are supported end-to-end across every client: `care_worker` and `manager`, each with a different navigation, permission set, and view of the data.

2. Development Timeline (Last 30 Days — Backend Repo)

Pulled directly from git history across all branches (main, main_deploy, main_deploy_updated_ui, feature/capacitor-android) on the Handover_Voice_Summarizer repository, 23 Jul – 21 Aug 2026, 21 commits:

Date	Work done
Jul 23	Introduced Redis and Celery for background processing; updated handover route, transcription service, conf test, test cases, and mobile-responsive UI.
Jul 24	Stability pass: fixed WebSocket auth, the Redis relay, Whisper model warmup, Celery concurrency, an audio retry bug, and added a Redis service to CI.
Jul 29	Nginx configuration changes.
Jul 30	Reworked email logic and summary/model logic with matching test-case updates; iterated twice on the CI pipeline.
Jul 31	Frontend and backend services changes.
Aug 3	Dashboard bug fix (on a fresh branch/commit reset).
Aug 5	Added the audit router; frontend changes; fixed a security bug and the summary/transcription flow.
Aug 6	Docker changes and test case updates.
Aug 10	Added manager ID handling and updated global search; fixed a resident issue, a handover issue, and related test cases (5 commits in one day).
Aug 13	Updated seed.py and shift logic, with matching test-case updates.
Aug 17	Updated celery_app and tasks.py; another CI pipeline update; on a separate branch, added a Capacitor Android setup.
Aug 21	New commit in progress (per local terminal output) — not yet reflected in the pushed history pulled for this report.

Two points worth flagging from the raw history:

- main_deploy and main_deploy_updated_ui are deployment-staging branches sitting behind main (last updated Aug 3 and Aug 6 respectively) — they haven't been synced with the audit logging, WebSocket, or assignment work merged into main since.
- feature/capacitor-android (Aug 17) is a separate, earlier exploration of wrapping the web frontend for Android via Capacitor. This looks like it predates — or ran in parallel with — the native Expo/React Native app covered in Section 6, which is the one now in active, more complete development.

The standalone mobile app (Expo/React Native) lives in its own repository, and no git history for it was available for this report — its section below is built from the current source tree only. Send its ``git log --all --pretty=format:"%ad | %s" --date=short --reverse`` output if you'd like the same day-by-day breakdown for it.

3. Architecture & Tech Stack

Backend

Layer	Technology
API framework	FastAPI 0.139, served via Gunicorn + Uvicorn workers

Database	PostgreSQL 16 + SQLAlchemy 2.0 (declarative) + Alembic (26 migrations to date)
Background jobs	Celery 5.3 (prefork pool) + Redis 7, plus Celery Beat for scheduled cleanup tasks
Real-time delivery	Native WebSocket endpoint, fanned out via a Redis pub/sub channel (ws_broadcast)
Speech-to-text	OpenAI Whisper (locally cached model, CPU-only Torch build)
Summarisation	Google Generative AI — Gemini 2.5 Flash
Auth & security	JWT auth, slowapi rate limiting, Cloudflare Turnstile CAPTCHA, audit-log middleware
Resilience	In-process circuit breaker around external calls (e.g. Gemini) with closed/open/half-open states
Email	Resend (OTP verification codes, urgent-handover manager alerts)

Frontend (Web)

React + Vite, with a shared component/page structure that renders both a full desktop layout and a dedicated responsive mobile layout (styles/mobile.css) from the same codebase — one deployment, two experiences.

Mobile App (native, new this period)

A separate Expo (SDK 54) / React Native 0.81 project, using expo-router for file-based navigation and expo-audio for in-app voice recording, sharing the same backend API and JWT auth as the web app.

Infrastructure

Docker Compose now runs six coordinated services: Postgres, Redis, a one-shot Alembic migration runner, the API (Gunicorn/Uvicorn), a Celery worker, Celery Beat, and an Nginx-served frontend reverse-proxying to the API — with health checks gating startup order across all of them.

4. Backend — What Shipped

- **Real-time notifications:** a WebSocket layer backed by Redis pub/sub now pushes live updates to managers — urgent handovers, resident discharges/deaths, and assignment changes all appear instantly rather than on next page load.
- **Resident – care worker assignment system:** a dedicated assignments module lets managers assign residents to specific care workers and vice-versa, with its own notification trail.
- **Audit logging:** an AuditLogMiddleware plus a dedicated audit router now record who changed what, with a queryable audit trail exposed to managers.
- **Email verification flow:** OTP-based email verification for new registrations (PendingUser table, Resend delivery, per-email and per-IP rate limits) is complete and stable.
- **Urgent-handover alerts wired up:** send_urgent_handover_email is now connected into the handover creation flow, so urgent cases trigger both an email and a real-time in-app notification to managers — this closes out an item that was previously parked.
- **Resilience:** a circuit breaker now protects calls to the Gemini API, and Celery concurrency was deliberately capped after a memory incident (see “Key Learnings”) to keep transcription stable under load.

- **Automated maintenance:** Celery Beat now runs scheduled cleanup of expired, never-verified pending registrations and auto clock-out for stale shifts.

5. Testing & CI/CD — What Shipped

The backend now has a genuine automated safety net rather than manual verification:

- 371 automated tests across auth, residents, shifts, handovers, assignments, notifications, audit, WebSocket, Celery integration, and general services.
- Around 94% statement coverage measured via pytest-cov, with a CI coverage floor enforced (build fails under 60%).
- GitHub Actions pipeline runs against real Postgres and Redis service containers, applies Alembic migrations, then runs the full suite with a coverage report artifact.
- Security scanning is built into CI: pip-audit for dependency vulnerabilities and bandit for static code analysis run on every push/PR.

This effectively completes the testing & CI/CD milestone that was previously on the roadmap as “not started.”

6. Web Frontend — What Shipped

The React web app now covers the full product surface for both roles, rendered from one codebase across desktop and mobile breakpoints:

Area	Pages / components
Core pages	Dashboard, Handovers, Residents, Shifts, Team, Profile (care worker & manager variants), Audit, Notifications, Auth
Handover workflow	New Handover modal, Handover detail modal, Handover card list with status/urgency badges
Resident & team management	Resident form/detail modals, care worker & manager form modals, assignment modal + chips
Platform UX	Global search, real-time WebSocket context, toast notifications, confirm dialogs, pagination, route guards, light/dark theme context, Turnstile hook, auto clock-out hook

The dedicated mobile.css stylesheet and mobile-specific components (BottomNav, etc.) keep the phone-browser experience purpose-built rather than a squeezed-down desktop layout.

7. Native Mobile App — New This Period

The standalone React Native/Expo app is the biggest addition since the last check-in. It has grown from a bare dashboard shell into a functionally complete companion app with 23 screens and 25 shared components, covering essentially the same feature set as the web app:

- Role-aware dashboard, activity/handover feed, and a dedicated handover detail screen with in-app voice recording (expo-audio).
- Resident list, resident detail, and resident create/edit forms.

- Team list, team-member detail, and manager detail screens, plus an assignment modal matching the web app's workflow.
- Notifications screen, global search, and a full audit log screen.
- Profile, edit-profile, change-password, and profile-form screens, including image picking for profile pictures.
- Secure JWT storage via expo-secure-store, themed UI with custom fonts (IBM Plex Sans/Mono, Source Serif), and haptic feedback.

The app talks to the same backend used by the web app (LAN IP through the Nginx proxy during development), and tunnel mode (Expo/ngrok) is the established way of testing it on-device without a same-network requirement.

8. Feature Coverage Snapshot

Capability	Backend	Web App	Mobile App
Voice recording & upload	✓	✓	✓
Whisper transcription	✓	—	—
Gemini structured summary	✓	—	—
Urgent-case email + real-time alert	✓	✓	Partial
Resident / care-worker assignment	✓	✓	✓
Audit trail	✓	✓	✓
Global search	✓	✓	✓
Automated tests	✓ (371 tests)	—	—

"Partial" = notification is delivered and visible on mobile, but push-notification delivery when the app is closed has not yet been confirmed end-to-end.

9. Open Items & Risks

- **Deployment hardening deferred:** CORS lockdown, backend-side Turnstile validation, HTTPS/TLS, Sentry error tracking, and object storage (S3/MinIO or Backblaze B2 as the no-card alternative) are all deliberately parked until a VPS is live — this is a scoping choice, not an oversight, but it is the main gap between “works” and “production-hardened.”
- **Resend domain:** still pending the care home providing a domain to send verified email from; OTP and alert emails currently rely on the interim setup.
- **Mobile push notifications:** in-app delivery works; background/closed-app push has not yet been verified.
- **Celery throughput ceiling:** worker concurrency is intentionally capped at 1 after an out-of-memory incident where two simultaneous Whisper model loads exhausted host memory. This trades throughput for stability until Docker Desktop's memory allocation is increased and the cap can be safely raised.
- **Bandit/pip-audit findings are non-blocking:** both scanners currently run with `| true` so CI doesn't fail on them yet — findings need triage before they can be made blocking.

10. Key Technical Learnings This Period

- Two Whisper model loads happening at once can OOM-kill a Celery worker and freeze the app for minutes — now mitigated by capping concurrency to 1 per worker.
- Running Alembic migrations as a separate one-shot Compose service (rather than on API container startup) stops a bad migration from crash-looping the API container.
- Splitting Celery Beat into its own service keeps the scheduler alive independently of worker restarts, and keeps the deliberately-constrained worker pool settings from being disturbed by unrelated changes.
- A circuit breaker around the Gemini call prevents a flaky external API from cascading into stuck Celery tasks.

11. Priorities for Next Period

- Finish VPS deployment: TLS, CORS hardening, Turnstile backend validation, Sentry, and object storage.
- Confirm mobile background push notifications end-to-end.
- Triage and resolve outstanding bandit/pip-audit findings, then make both checks blocking in CI.
- Secure a Resend-verified sending domain from the care home.
- Re-evaluate the Celery concurrency cap once host memory allocation is increased.

Report generated from a review of the project's backend, web frontend, and mobile app source trees.